

# AUSTIN REILLEY BENSON

*Quant finance researcher*  
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<https://arbenson.github.io>

<https://scholar.google.com/citations?user=Bz0qNoQAAAAJ&hl=en>

## *Professional experience*

|                                                                 |           |
|-----------------------------------------------------------------|-----------|
| D. E. Shaw & Co., Equities Group                                | 2021–2025 |
| Vice President                                                  | 2024–2025 |
| Quant Researcher                                                | 2021–2023 |
| Cornell University, Department of Computer Science              | 2017–2021 |
| Assistant Professor                                             | 2018–2021 |
| Field Memberships: Center for Applied Mathematics; Data Science |           |
| Postdoctoral Associate                                          | 2017–2018 |
| Internships                                                     |           |
| Research Intern, Google                                         | 2016      |
| Research Intern, Google                                         | 2015      |
| Research Associate, Sandia National Laboratories                | 2014      |
| Research Intern, HP Labs                                        | 2013      |
| Software Engineering Intern, Google                             | 2012      |
| Software Engineering Intern, Google                             | 2011      |

## *Education*

|                                                         |           |
|---------------------------------------------------------|-----------|
| Stanford University                                     | 2012–2017 |
| PhD, Computational and Mathematical Engineering         |           |
| MS, Computational and Mathematical Engineering          |           |
| University of California, Berkeley                      | 2008–2012 |
| BS, Electrical Engineering and Computer Sciences (EECS) |           |
| BA, Applied Mathematics                                 |           |

## ***Selected publications*** (complete list at end of CV)

1. Higher-order organization of complex networks.  
Austin R. Benson, David F. Gleich, and Jure Leskovec.  
*Science*, 2016.
2. Simplicial closure and higher-order link prediction.  
Austin R. Benson, Rediet Abebe, Michael T. Schaub, Ali Jadbabaie, and Jon Kleinberg.  
*Proceedings of the National Academy of Sciences (PNAS)*, 2018.
3. A Unifying Generative Model for Graph Learning Algorithms: Label Propagation, Graph Convolutions, and Combinations.  
Junteng Jia and Austin R. Benson.  
*SIAM Journal on Mathematics of Data Science (SIMODS)*, 2022.
4. Combining Label Propagation and Simple Models Out-performs Graph Neural Networks.  
Qian Huang, Horace He, Abhay Singh, Ser-Nam Lim, and Austin R. Benson.  
*Proceedings of the International Conference on Learning Representations (ICLR)*, 2021.
5. Choosing to grow a graph: Modeling network formation as discrete choice.  
Jan Overgoor, Austin R. Benson, and Johan Ugander.  
*Proceedings of the Web Conference (WWW)*, 2019.
6. Random Walks on Simplicial Complexes and the normalized Hodge 1-Laplacian.  
Michael T. Schaub, Austin R. Benson, Paul Horn, Gabor Lippner, and Ali Jadbabaie.  
*SIAM Review*, 2020.
7. Hypergraph Cuts with General Splitting Functions.  
Nate Veldt, Austin R. Benson, and Jon Kleinberg.  
*SIAM Review*, 2022.
8. On the relevance of irrelevant alternatives.  
Austin R. Benson, Ravi Kumar, and Andrew Tomkins.  
*Proceedings of the International World Wide Web Conference (WWW)*, 2016.
9. The spacey random walk: a stochastic process for higher-order data.  
Austin R. Benson, David F. Gleich, and Lek-Heng Lim.  
*SIAM Review*, 2017.
10. Motifs in temporal networks.  
Ashwin Paranjape, Austin R. Benson, and Jure Leskovec.  
*Proceedings of the International Conference on Web Search and Data Mining (WSDM)*, 2017.

## ***Academic awards***

|                                                                        |      |
|------------------------------------------------------------------------|------|
| NSF Faculty Early Career Development Program (CAREER) Award            | 2021 |
| Kavli Fellow, Kavli Frontiers of Science, National Academy of Sciences | 2020 |
| JPMorgan Chase Faculty AI Research Award                               | 2020 |
| Best Research Paper Award, KDD '19                                     | 2019 |
| Best Research Paper Award Runner-up, ASONAM '19                        | 2019 |
| LAA Early Career Speaker, International Linear Algebra Society         | 2019 |
| Outstanding program committee member, WSDM '19                         | 2019 |
| Stanford Gene Golub Doctoral Dissertation Award                        | 2017 |
| Teaching Fellow, ICME, Stanford University                             | 2016 |
| Office of Technology Licensing Stanford Graduate Fellowship            | 2012 |

## ***Mentorship and advising***

*Students, postdocs, and interns from time at Cornell and DE Shaw. Next career stage indicated in parentheses.*

### ***Postdocs***

Jonas Juul (Assistant Professor, IT University of Copenhagen, Computer Science Department)  
Nate Veldt (Assistant Professor, Texas A&M, Department of Computer Science and Engineering)

### ***PhD Students***

Kiran Tomlinson (MSR)  
Ilya Amburg (PNNL)  
Junteng Jia (Meta)

### ***Undergraduate Students***

Qian Huang (MIT PhD)  
*Finalist, CRA Outstanding Undergraduate Researcher Award, 2020 & 2021*  
*Recipient, Computer Science Department Prize for Academic Excellence*  
Abhay Singh (Citadel GQS)  
Horace He (Meta)  
*Honorable Mention, CRA Outstanding Undergraduate Researcher Award, 2020*  
Derek Lim (MIT PhD)  
*Honorable Mention, CRA Outstanding Undergraduate Researcher Award, 2021*  
Leah Ajmani (Minnesota PhD)  
Shangdi Yu (MIT PhD)  
*Recipient, Computer Science Department Prize for Academic Excellence*  
Xiang (Felix) Fu (MIT PhD)

### ***Summer Interns***

Jennifer Sun (DE Shaw)  
Raghav Somani (DE Shaw)  
Feicheng Wang (Citadel Securities)

## ***Complete list of papers and publications***

76. Graph neural network modeling of grain-scale anisotropic elastic behavior using simulated and measured microscale data.  
Darren C. Pagan, Calvin R. Pash, Austin R. Benson, and Matthew P. Kasemer.  
*npj Computational Materials*, 2024.
75. Hypergraph patterns and collaboration structure.  
Jonas L. Juul, Austin R. Benson, and Jon Kleinberg.  
*Frontiers in Physics*, 2024.
74. Graph-based methods for discrete choice.  
Kiran Tomlinson and Austin R. Benson.  
*Network Science*, 2023.
73. A Unifying Generative Model for Graph Learning Algorithms: Label Propagation, Graph Convolutions, and Combinations.  
Junteng Jia and Austin R. Benson.  
*SIAM Journal on Mathematics of Data Science (SIMODS)*, 2022.
72. Nonlinear Feature Diffusion on Hypergraphs.  
Francesco Tudisco, Konstantin Prokopchik, and Austin R. Benson.  
*Proceedings of the International Conference on Machine Learning (ICML)*, 2022.
71. fauci-email: a JSON Digest of Anthony Fauci’s Released Emails.  
Austin R. Benson, Nate Veldt, and David F. Gleich.  
*Proceedings of the International AAAI Conference on Web and Social Media (ICWSM)*, 2022.
70. Understanding non-linearity in graph neural networks from the Bayesian-inference perspective.  
Rongzhe Wei, Haoteng Yin, Junteng Jia, Austin R. Benson, and Pan Li.  
*Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
69. Combinatorial characterizations and impossibilities for higher-order homophily.  
Nate Veldt, Austin R. Benson, and Jon Kleinberg.  
*Science Advances*, 2023.
68. Diverse and Experienced Group Discovery via Hypergraph Clustering.  
Ilya Amburg, Nate Veldt, and Austin R. Benson.  
*SIAM International Conference on Data Mining (SDM)*, 2022.
67. Augmented Sparsifiers for Generalized Hypergraph Cuts.  
Austin R. Benson, Jon Kleinberg, and Nate Veldt.  
*Journal of Machine Learning Research (JMLR)*, 2023.
66. Hypergraph Cuts with General Splitting Functions.  
Nate Veldt, Austin R. Benson, and Jon Kleinberg.  
*SIAM Review*, 2022.
65. Approximate Decomposable Submodular Function Minimization for Cardinality-Based Components.  
Austin R. Benson, Jon Kleinberg, and Nate Veldt.  
*Advances in Neural Information Processing Systems (NeurIPS)*, 2021.
64. Communication-efficient distributed eigenspace estimation.  
Vasileios Charisopoulos, Austin R. Benson, and Anil Damle.  
*SIAM Journal on Mathematics of Data Science (SIMODS)*, 2021.
63. Hypergraph clustering: from blockmodels to modularity.  
Philip S. Chodrow, Nate Veldt, and Austin R. Benson.  
*Science Advances*, 2021.

62. The Generalized Mean Densest Subgraph Problem.  
Nate Veldt, Austin R. Benson, and Jon Kleinberg.  
*Proceedings of the International Conference on Knowledge Discovery & Data Mining (KDD)*, 2021.
61. Choice Set Confounding in Discrete Choice.  
Kiran Tomlinson, Johan Ugander, and Austin R. Benson.  
*Proceedings of the International Conference on Knowledge Discovery & Data Mining (KDD)*, 2021.
60. Learning Interpretable Feature Context Effects in Discrete Choice.  
Kiran Tomlinson and Austin R. Benson.  
*Proceedings of the International Conference on Knowledge Discovery & Data Mining (KDD)*, 2021.
59. Expertise and Dynamics within Crowdsourced Musical Knowledge Curation: A Case Study of the Genius Platform.  
Derek Lim and Austin R. Benson.  
*Proceedings of the International AAAI Conference on Web and Social Media (ICWSM)*, 2021.
58. Higher-order Network Analysis Takes Off, Fueled by Old Ideas and New Data.  
Austin R. Benson, David F. Gleich, and Desmond J. Higham.  
*SIAM News*, 2021 (online).
57. Combining Label Propagation and Simple Models Out-performs Graph Neural Networks.  
Qian Huang, Horace He, Abhay Singh, Ser-Nam Lim, and Austin R. Benson.  
*Proceedings of the International Conference on Learning Representations (ICLR)*, 2021.
56. Planted Hitting Set Recovery in Hypergraphs.  
Ilya Amburg, Jon Kleinberg, and Austin R. Benson.  
*Journal of Physics: Complexity*, 2021.
55. Nonlinear Higher-Order Label Spreading.  
Francesco Tudisco, Austin R. Benson, and Konstantin Prokopchik.  
*Proceedings of the Web Conference (WWW)*, 2021.
54. Random Graphs with Prescribed K-Core Sequences: A New Null Model for Network Analysis.  
Katherine Van Koeveering, Austin R. Benson, and Jon Kleinberg.  
*Proceedings of the Web Conference (WWW)*, 2021.
53. Capturing Tie Strength with Algebraic Topology.  
Arnab Sarker, Jean-Baptiste Seby, Austin R. Benson, and Ali Jadbabaie.  
arXiv:2108.02091, 2021.
52. Edge Proposal Sets for Link Prediction.  
Abhay Singh, Qian Huang, Sijia Linda Huang, Omkar Bhalerao, Horace He, Ser-Nam Lim, and Austin R. Benson.  
arXiv:2106.15810, 2021.
51. Graph Belief Propagation Networks.  
Junteng Jia, Cenk Baykal, Vamsi K. Potluru, and Austin R. Benson.  
arXiv:2106.03033, 2021.
50. Better Set Representations For Relational Reasoning.  
Qian Huang, Horace He, Abhay Singh, Yan Zhang, Ser-Nam Lim, and Austin R. Benson.  
*Advances in Neural Information Processing Systems (NeurIPS)*, 2020.
49. Entrywise convergence of iterative methods for eigenproblems.  
Vasileios Charisopoulos, Austin R. Benson, Anil Damle.  
*Advances in Neural Information Processing Systems (NeurIPS)*, 2020.

48. Residual Correlation in Graph Neural Network Regression.  
Junteng Jia and Austin R. Benson.  
*Proceedings of the International Conference on Knowledge Discovery & Data Mining (KDD)*, 2020.
47. Minimizing Localized Ratio Cut Objectives in Hypergraphs.  
Nate Veldt, Austin R. Benson, and Jon Kleinberg.  
*Proceedings of the International Conference on Knowledge Discovery & Data Mining (KDD)*, 2020.
46. Choice Set Optimization Under Discrete Choice Models of Group Decisions.  
Kiran Tomlinson and Austin R. Benson.  
*Proceedings of the International Conference on Machine Learning (ICML)*, 2020.
45. Neighborhood and PageRank methods for pairwise link prediction.  
Huda Nassar, Austin Benson, and David F. Gleich.  
*Social Network Analysis and Mining (SNAM)*, 2020.
44. Network Interpolation.  
Thomas Reeves, Anil Damle, and Austin R. Benson.  
*SIAM Journal on Mathematics of Data Science (SIMODS)*, 2020.
43. Measuring Directed Triadic Closure with Closure Coefficients.  
Hao Yin, Austin R. Benson, and Johan Ugander.  
*Network Science*, 2020.
42. Random Walks on Simplicial Complexes and the normalized Hodge 1-Laplacian.  
Michael T. Schaub, Austin R. Benson, Paul Horn, Gabor Lippner, and Ali Jadbabaie.  
*SIAM Review*, 2020.
41. Clustering in graphs and hypergraphs with categorical edge labels.  
Ilya Amburg, Nate Veldt, and Austin R. Benson.  
*Proceedings of the Web Conference (WWW)*, 2020.
40. Frozen Binomials on the Web: Word Ordering and Language Conventions in Online Text.  
Katherine Van Koeveering, Austin R. Benson and Jon Kleinberg.  
*Proceedings of the Web Conference (WWW)*, 2020.
39. Using cliques with higher-order spectral embeddings improves graph visualizations.  
Huda Nassar, David Gleich, Austin R. Benson, Shweta Jain and Caitlin Kennedy.  
*Proceedings of the Web Conference (WWW)*, 2020.
38. Retrieving Top Weighted Triangles in Graphs.  
Raunak Kumar, Paul Liu, Moses Charikar, and Austin R. Benson.  
*Proceedings of the International Conference on Web Search and Data Mining (WSDM)*, 2020.
37. Over-parametrized neural networks as under-determined linear systems.  
Austin R. Benson, Anil Damle, and Alex Townsend.  
arXiv:2010.15959, 2020.
36. Neural Jump Stochastic Differential Equations.  
Junteng Jia and Austin R. Benson.  
*Advances in Neural Information Processing Systems (NeurIPS)*, 2019.
35. Modeling and Analysis of Tagging Networks in Stack Exchange Communities.  
Xiang Fu, Shangdi Yu, and Austin R. Benson.  
*Journal of Complex Networks*, 2019.
34. Unsupervised learning of dislocation motion.  
Darren C. Pagan, Thien Q. Phan, Jordan S. Weaver, Austin R. Benson, and Armand J. Beaudoin.  
*Acta Materialia*, 2019.

33. Automated Grain Yield Behavior Classification.  
Darren C. Pagan, Jakob Kaminsky, Wesley A. Tayon, Kelly E. Nygren, Armand J. Beaudoin, and Austin R. Benson.  
*The Journal of The Minerals, Metals & Materials Society (JOM)*, 2019.
32. Computing Tensor Z-eigenvectors with Dynamical Systems.  
Austin R. Benson and David F. Gleich.  
*SIAM Journal on Matrix Analysis and Applications (SIMAX)*, 2019.
31. Pairwise Link Prediction.  
Huda Nassar, Austin R. Benson, and David F. Gleich.  
*Proceedings of the International Conference on Advances in Social Networks Analysis and Mining (ASONAM)*, 2019.
30. Graph-based Semi-Supervised & Active Learning for Edge Flows.  
Junteng Jia, Michael T. Schaub, Santiago Segarra, and Austin R. Benson.  
*Proceedings of the International Conference on Knowledge Discovery & Data Mining (KDD)*, 2019.
29. Network Density of States.  
Kun Dong, Austin R. Benson, and David Bindel.  
*Proceedings of the International Conference on Knowledge Discovery & Data Mining (KDD)*, 2019.
28. Three hypergraph eigenvector centralities.  
Austin R. Benson.  
*SIAM Journal on Mathematics of Data Science (SIMODS)*, 2019.
27. Link Prediction in Networks with Core-Fringe Data.  
Austin R. Benson and Jon Kleinberg.  
*Proceedings of the Web Conference (WWW)*, 2019.
26. Choosing to grow a graph: Modeling network formation as discrete choice.  
Jan Overgoor, Austin R. Benson, and Johan Ugander.  
*Proceedings of the Web Conference (WWW)*, 2019.
25. Random Spatial Network Models with Core-Periphery Structure.  
Junteng Jia and Austin R. Benson.  
*Proceedings of the International Conference on Web Search and Data Mining (WSDM)*, 2019.
24. Sampling Methods for Counting Temporal Motifs.  
Paul Liu, Austin R. Benson, and Moses Charikar.  
*Proceedings of the International Conference on Web Search and Data Mining (WSDM)*, 2019.
23. The Local Closure Coefficient: A New Perspective On Network Clustering.  
Hao Yin, Austin R. Benson, and Jure Leskovec.  
*Proceedings of the International Conference on Web Search and Data Mining (WSDM)*, 2019.
22. Incrementally Updated Spectral Embeddings.  
Vasileios Charisopoulos, Austin R. Benson, and Anil Damle.  
arXiv:1909.01188, 2019.
21. Simplicial closure and higher-order link prediction.  
Austin R. Benson, Rediet Abebe, Michael T. Schaub, Ali Jadbabaie, and Jon Kleinberg.  
*Proceedings of the National Academy of Sciences (PNAS)*, 2018.
20. Found Graph Data and Planted Vertex Covers.  
Austin R. Benson and Jon Kleinberg.  
*Advances in Neural Information Processing Systems (NeurIPS)*, 2018.

19. Sequences of Sets.  
Austin R. Benson, Ravi Kumar, and Andrew Tomkins.  
*Proceedings of the International Conference on Knowledge Discovery & Data Mining (KDD)*, 2018.
18. Higher-order clustering in networks.  
Hao Yin, Austin R. Benson, and Jure Leskovec.  
*Physical Review E*, 2018.
17. A discrete choice model for subset selection.  
Austin R. Benson, Ravi Kumar, and Andrew Tomkins.  
*Proceedings of the International Conference on Web Search and Data Mining (WSDM)*, 2018.
16. Local higher-order graph clustering.  
Hao Yin, Austin R. Benson, Jure Leskovec, and David F. Gleich.  
*Proceedings of the International Conference on Knowledge Discovery & Data Mining (KDD)*, 2017.
15. Motifs in temporal networks.  
Ashwin Paranjape, Austin R. Benson, and Jure Leskovec.  
*Proceedings of the International Conference on Web Search and Data Mining (WSDM)*, 2017.
14. The spacey random walk: a stochastic process for higher-order data.  
Austin R. Benson, David F. Gleich, and Lek-Heng Lim.  
*SIAM Review*, 2017.
13. Higher-order organization of complex networks.  
Austin R. Benson, David F. Gleich, and Jure Leskovec.  
*Science*, 2016.
12. General tensor spectral co-clustering for higher-order data.  
Tao Wu, Austin R. Benson, and David F. Gleich.  
*Advances in Neural Information Processing Systems (NeurIPS)*, 2016.
11. Modeling user consumption sequences.  
Austin R. Benson, Ravi Kumar, and Andrew Tomkins.  
*Proceedings of the International World Wide Web Conference (WWW)*, 2016.
10. On the relevance of irrelevant alternatives.  
Austin R. Benson, Ravi Kumar, and Andrew Tomkins.  
*Proceedings of the International World Wide Web Conference (WWW)*, 2016.
9. Improving the numerical stability of fast matrix multiplication.  
Grey Ballard, Austin R. Benson, Alex Druinsky, Benjamin Lipshitz, and Oded Schwartz.  
*SIAM Journal on Matrix Analysis and Applications (SIMAX)*, 2016.
8. Tensor spectral clustering for partitioning higher-order network structures.  
Austin R. Benson, David F. Gleich, and Jure Leskovec.  
*Proceedings of the SIAM International Conference on Data Mining (SDM)*, 2015.
7. A framework for practical parallel fast matrix multiplication.  
Austin R. Benson and Grey Ballard.  
*Proceedings of the Symposium on Principles and Practice of Parallel Programming (PPoPP)*, 2015.
6. Scalable methods for nonnegative matrix factorizations of near-separable tall-and-skinny matrices.  
Austin R. Benson, Jason D. Lee, Bartek Rajwa, and David F. Gleich.  
*Advances in Neural Information Processing Systems (NeurIPS)*, 2014.
5. Learning multifractal structure in large networks.  
Austin R. Benson, Carlos Riquelme, and Sven Schmit.  
*Proceedings of the International Conference on Knowledge Discovery & Data Mining (KDD)*, 2014.



4. A parallel directional Fast Multipole Method.  
Austin R. Benson, Jack Poulson, Kenneth Tran, Björn Engquist, and Lexing Ying.  
*SIAM Journal on Scientific Computing (SISC)*, 2014.
3. Silent error detection in numerical time-stepping schemes.  
Austin R. Benson, Sven Schmit, and Robert Schreiber.  
*International Journal of High Performance Computing Applications (IJHPCA)*, 2014.
2. Direct QR factorizations for tall-and-skinny matrices in MapReduce architectures.  
Austin R. Benson, David F. Gleich, and James Demmel.  
*Proceedings of the IEEE International Conference on Big Data (BigData)*, 2013.
1. The Gamma-Ray Imaging Framework.  
Austin R. Benson, Mark S. Bandstra, Daniel H. Chivers, Timothy Aucott, Ben Augarten, Cameron Bates, Adam Midvidy, Ryan Pavlovsky, James Siegrist, Kai Vetter, and Ben Yee.  
*IEEE Transactions on Nuclear Science*, 2013.